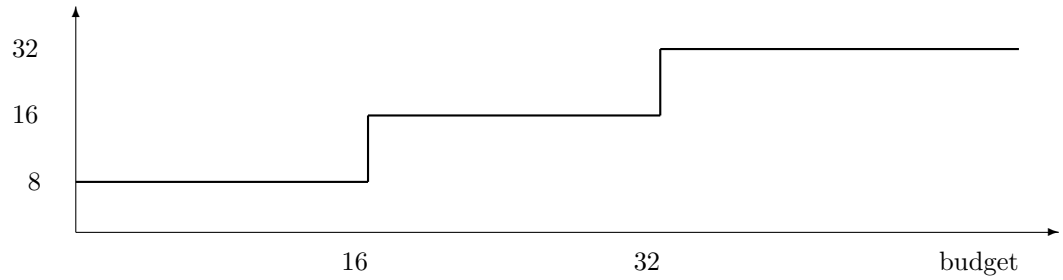
Independent GLV table-selector review

2026-11-07. Eight named theorems compile under pinned Lean 4.34.1. For the same nonnegative input and positive log factor, changing the budget divisor from 48 to 12 cannot decrease the selected 8/16/32 table size. This is a conditional selector-arithmetic result.

Budget	C loop	Lean formula
$b < 16$	8	8
$16 \le b < 32$	16	16
$b \ge 32$	32	32

The loop starts at 8, doubles under both the budget and cap-32 guards, and changes at most twice. Positive denominators give $b_{48} \le b_{12}$; the case table is nondecreasing. A third step is fixed. **Checks.** Fresh `lean TableBudget.lean` exited zero; each of eight printed axiom sets contains only Lean's `propext` and possibly `Quot.sound`. The live C-source binding checker passed and rejected wrong-divisor and wrong-cap mutations. A blind algebra reviewer and a separate source-fidelity reviewer concurred. Exact files, hashes, and commands are in the companion Markdown and machine receipts. **Boundary.** The C branch implies a positive log factor. The external `c.m>0` invariant is required for C division and was not proved here. The review does not replay the experiment's performance rows. An older source hash in `LEAN_PROOF.md` is stale; the pinned live C hash and current target agree. **Diagram.** The vector source `diagram.svg` plots the exact three-case selector. It represents proved arithmetic, not measured operation counts.



Source target: `927d1ead...`. C SHA-256: `60796724...afe20`. Full receipt and reviewer attestations accompany this PDF.